

HW # 14

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Problem 1:

To show R_θ is unitary,
 $R_\theta R_\theta^* = I$

$$\begin{aligned} R_\theta R_\theta^* &= \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2\theta + \sin^2\theta & 0 \\ 0 & \sin^2\theta + \cos^2\theta \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ &= I \end{aligned}$$

Therefore, R_θ is unitary.

Problem 2:

$$F_U = 2UU^T - I \quad \text{if } U \text{ is a unit matrix}$$

$$F_U F_U^T = I$$

$$\begin{aligned} F_U F_U^T &= (2UU^T - I)(2UU^T - I)^T \\ \because (A+B)^T &= A^T + B^T \\ \because (AB)^T &= B^T A^T \\ &= (2UU^T - I)((2UU^T)^T - (I)^T) \\ &= (2UU^T - I)(2U^*U^T - I) \\ &= 4UU^T UU^T - 4UU^T + I \\ &= 4UU^T - 4UU^T + I \\ &= I \end{aligned}$$

Proved.